Operation Bulletin

Breakdown of sewing operations and sequencing them are critically important in an Assembly Line system because the construction of a garment is planned in steps where different operators are responsible for sewing different parts of the same garment. Wherein each and every machining and non-machining operation are identified and listed in a flow chart (Binran, 1994) or process flow chart (StitchWorld, 2008), showing the precedence relationship between the operations. For every operation, there is some necessary information listed in the process flow charts like operation name or operation code, machine/equipment used and work content of the operation. However there are many more operation parameters that need to be conveyed to sewing supervisor and listing everything within the chart is always not possible (will make the chart clumsy or too big). Therefore, the operations are also listed in a separate tabular form mentioning different sewing parameters in different columns against each operation. The tabular format is referred as "Quality Standard Spec Sheet" (Solinger, 1986) or commonly operation bulletin (StitchWorld, 2008).

Every garment is broken down into different sewing and non-sewing operation keeping in mind two things; the cycle time of the operation should be short enough to be repeatedly performed and hand motions can be easily standardized. Secondly, the operation should have a visibly clear start and stop action (with or without back tack and except for a burst in a single operation).

For example, collar making in men's shirts can be divided into collar runstitch, collar turn inside out collar pressing, collar topstitch, collar center trimming and notching, band hemming, band and collar joining, and turning the band to finish the collar. These operations determine the machinery and equipment required for each operation. There seems to be a practice of laying more emphasis on sewing operations and ignoring non-sewing operations, but both are equally important since a bottleneck in either operation will adversely affect the preceding or succeeding operation.

PREPARING A PROCESS FLOW CHART

Symbols are used to create process flow charts to prefix the statements identifying each operation. The symbols enable one to classify quickly the category of each operation (i.e. operation, transportation, storage, inspection, delay and activity outside the scope of investigation). Solinger argued that a process flow chart is an inadequate tool for production planning purposes because it is a diagram without any time or space scales (Solinger, 1986). He had suggested "Flow Process Grid", which is a two-dimensional graph where Y-axis is the timeline of the production system and the longitudinal space line of a plant layout. The X-axis depicts the lateral space relationship among the work and temporary storage stations. The Figure 9.1 shows the process flow chart and Table 9.1 shows the flow process grid. Another scientific way of representing the sewing flow process of any garment is network diagram figure 9.2 (Colovic, 2011) or PERT network (Jana, 2011), which shows the precedence relationship of the operations, the work content of each operation and the critical path. The critical path indicates the throughput time once the garment is manufactured in Progressive Bundle Unit (PBU) system. The process flow chart and flow process grid are drawn vertically; however the network diagram is drawn horizontally. The sewing machine catalogues, training institutes, and the industry commonly use process flow chart.

PREPARING AN OPERATION BULLETIN

The Operation Bulletin (OB) comprises of the operation name and description in one column and the machine and operation parameters in the other. Although no definitive format of the OB is available, Binran (Binran, 1994) lists six different parameters namely sketch of operation, operation number, description of operation, seam diagram, type of machine and operation time. Quality Standard Spec Sheet (Solinger, 1986) lists fifteen different parameters namely stitch type, SPI, stitch width, needle thread size, needle thread tension, bobbin/looper thread size, needle size, seam type, seam sketch, machine type, presser foot, feed type and size, throat plate, maximum machine RPM and attachments. Some organizations prepare OB with all possible parameters of the operations while other companies work with OB listing the most important operation parameters. While preparing the OB, some information like stitch and seam type, stitches per inch are extracted from the techpack while IE suggests the other details. The techpack lists out what all is required/expected (output parameters) in the sample, and accordingly the OB lists out all the process parameters (input parameters) to achieve the required output. Keeping

Table 9.1 Complete Flow Process Grid for a Jeans Style (Adapted from Solinger, 1986)

Operation Level	Operation Title	Symbol for Machine Used	Minutes per Unit	Minutes per bundle of 20	100% Operators Needed per Hour	Operation Title	Symbol for Machine Used	Minutes per Unit	Minutes per bundle of 20	100% Operators Needed per Hour	Operation Title	Symbol for Machine Used	Minutes per Unit	Minutes per bundle of 20	100% Operators Needed per Hour
21	Clean and Inspect	LM	2.21	44.2	5.53										
20	Attach Buttons	BP	0.93	18.6	2.33										
19	Sew Button hole	O	0.23	4.6	0.58										
18	Attach Ticket	CR	0.37	7.4	0.93										
17	Bartack Fly	A	0.39	7.8	0.98										
16	Bartack Pockets	A	0.40	8	1.00										
15	Bartack Belt Loops	A	1.51	30.2	3.78										
14	Close Wasit Band Ends	CR	0.85	17	2.13	Cut Belt Loops	CP	0.10	2.0	0.25					
13	Cut Waist Band Ends	M	0.57	11.4	1.43	Sew Belt Loops	Re	0.17	3.4	0.42					

12	Atach Waist Band	PJ	1.05	21	2.63										
11	Hem Leg Bottom	CR	2.01	40.2	5.03										
10	Close Inside Leg	RS	0.86	17.2	2.15										
9	Close Sides	T	1.66	33.2	4.15										
8	Join Fronts	CR	0.39	7.8	0.98	Join Backs	T	0.65	13.0	1.62					
7	Attach Right Fly and Zipper	CR	0.54	10.8	1.35	Attach Tab Holders	CR	3.01	60.2	7.52					
6	Attach Zipper to Left Fly	DR	0.67	13.4	1.68	Press Tab Holders	M	0.20	4.0	0.50	Buttonhole Tabs	0	0.46	9.2	1.15
5	Close Front Pocket Pouch	RS	0.78	15.6	1.95	Attach Back Pockets	DR	1.42	28.4	3.55	Topstitch Tabs	DR	1.02	20.4	2.55
4	Tack Front Pockets	CR	0.81	16.2	2.03	Press Back Pockets	M	1.97	39.4	4.92	Turn & Trim Tabs	M	0.78	15.6	1.95
3	Attach Front Pockets	DR	0.73	14.6	1.83						Sew Back Tabs	CR	0.78	15.6	1.95
2	Attach Left Fly and Topstitch	CR	0.64	12.8	1.60	Attach Facing to Pouch	CR	0.47	9.4	1.17					
1	Overcast Flys	RD	0.25	5.0	0.63	Overcast Facing	RD	0.25	5.0	0.62					

- *Target - 150 pairs of Jeans per hour*
- *Total Time per Unit = 29.13 minutes (Total 34 operations arranged in 21 operation levels)*
- *Maximum Total Level Time per bundle of 20 = 448.6 minutes = 7.48 hours (Adding all the grey coloured cells)*
- *Total waiting Time per Bundle between Levels = 582.6 minutes = 9.71 hours (29.13 multiplying by bundle size of 20)*
- *Total Inventory-In-Process Time per Bundle (20) = 17.19 hours*
- *Total 100% Operators needed for achieving the target = 72.80*

Figure 9.1: Process Flow Chart

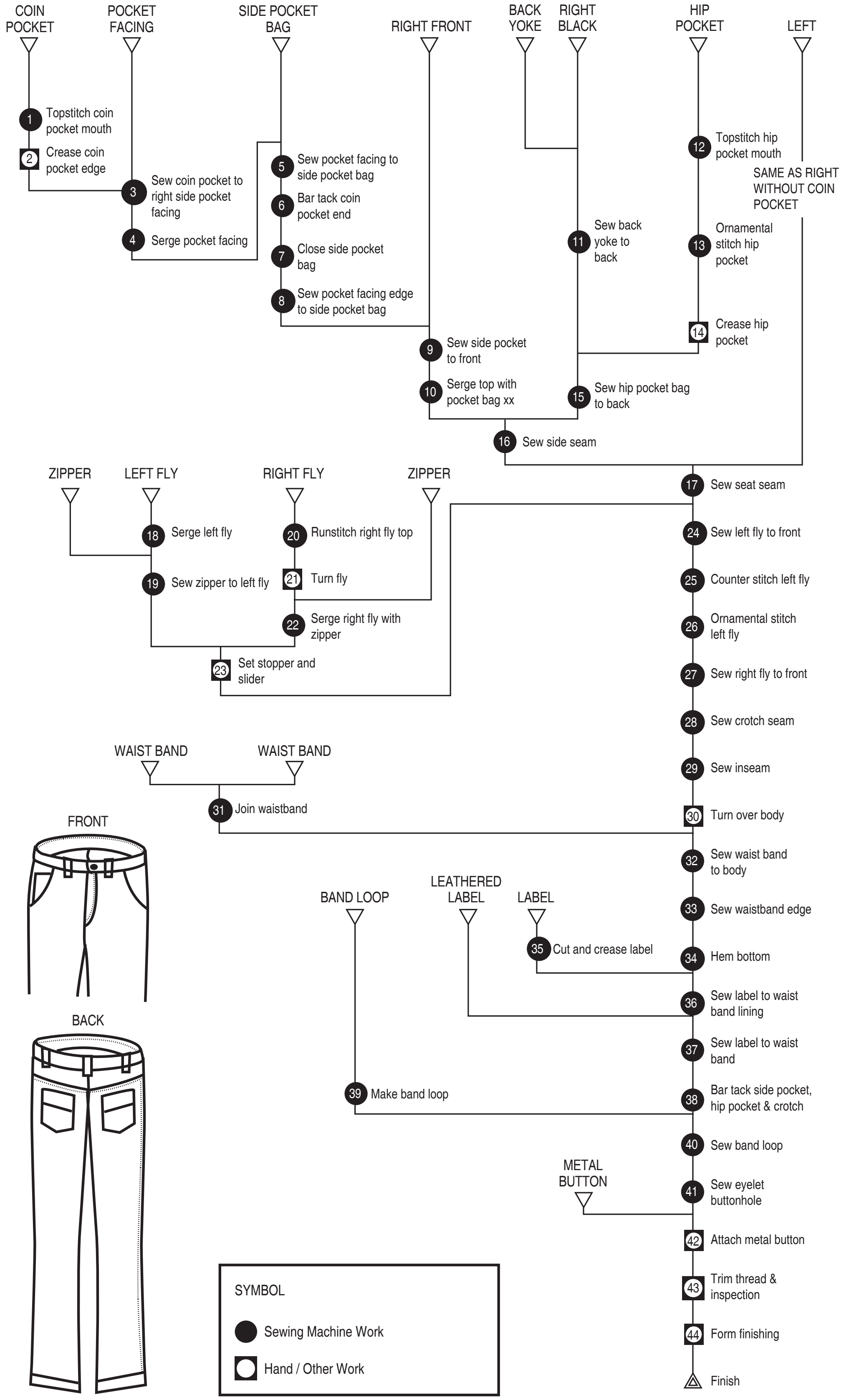
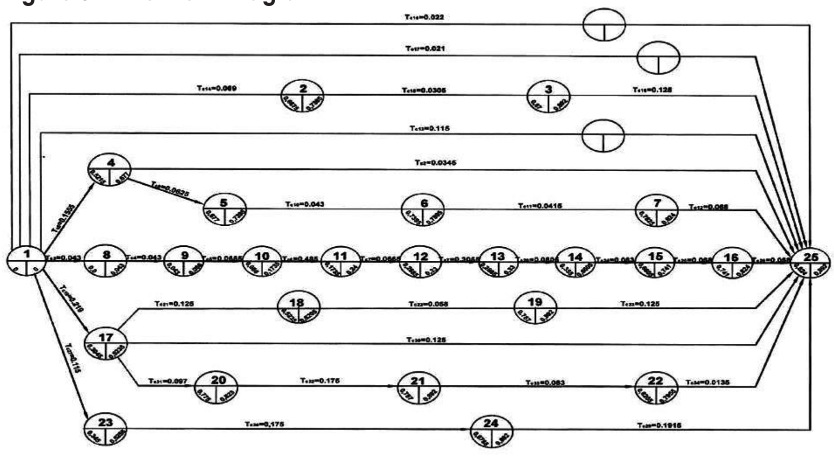


Figure 9.2: Network Diagram



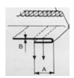
in sync with industry practices the Operation Bulletin (OB) (Table 9.2) shows a comprehensive list of 23 parameters wherein the importance and usefulness of each parameter is explained separately.

MACHINE SELECTION

The sewing machine has three distinct features that characterize the bed type, stitch type, and feed type, but it is a common practice in the industry while preparing the OB to make a single column giving either the generic specifications like 'flatbed SNLS' or the machine brand with its model number (like Juki DDL 9000). However, while writing in a single column, one tends to miss some crucial information like the feed type or bed type; so it is important that all essential information be provided in separate columns. If the organization is large and has a centralized IE preparing the OB, it is quite likely that the machine brand and model number details may not be known to the IE; in this case, the generic specification is acceptable. But in a smaller organization, where the machine brand and model number is available, the IE should prepare the OB by writing the machine's model number in a single column, which would automatically indicate the bed, stitch, and feed type, along with any other additional features. For example, JUKI DLM 5400N indicates that it is a flatbed lockstitch drop feed machine with vertical edge trimmer. It is advisable that engineers of the IE department should be knowledgeable with all machine types and their features, and should mention the appropriate machine model number for each operation.

It was often noticed that appropriate machines were not used as suggested by the IE, resulting in additional operation(s) and increase in SAM value. For example, if IE has recommended JUKI DLM 5400N machine

Table 9.2: Sample Operation Bulletin

Operation name	Stitch type	Bed type	Feed type	Machine's additional features	Presser foot type	Attachments	Work aid	Needle type	Needle point	Needle size	Seam diagram	Stitches per inch	Needle thread type	Needle thread Tex/ Ticket No.	Bobbin/ Looper thread type	Bobbin/ Looper thread Tex/ Ticket No.	Seam length in cm	Total thread consumption	SAM	Expected target/ hours	No. of M/C required
Men's shirt collar runstitch	301	Flatbed	Drop feed	SNLS with vertical edge trimmer	Solid	NIL	Step rack for loading	DB x 1	Groz-Beckert R	75/11		12	Poly poly core spun	30/100	Poly poly core spun	30/100	54 cm	54 cm x 2.54 cm	0.644	100	1.07
Jeans inseam	401	Feed-of-the-arm	Drop feed	Rubber teethed roller puller feed	Hinged	Lap seam folder	NIL	UY 128 GAS	Groz-Beckert RG	110/18		10	Poly cotton core spun	105/30	Poly cotton core spun	60/50	156 cm	156 cm	0.65	60	0.65
T-shirt bottom hem	406	Raised bed	Differential feed	Left hand under fabric trimmer	Hinged	Down turn feller	NIL	UY 128 GAS	Groz-Beckert FFG/SES	75/11		12	Coats seamsoft	24/140	Coats seamsoft	18/160	145 cm	145 cm	0.17	200	0.56
Collar turn inside out and blocking	NIL	NIL	NIL	NS-44 Collar turning & blocking	NIL	NIL	Rack	NIL	NIL	NIL	NIL	NIL	NIL	NIL	NIL	NIL	NIL	NIL	0.47	100	0.78

for collar run stitch operation, then the collar edge is simultaneously trimmed during the run stitch and can be taken for collar turn inside out operation. But, suppose the factory does not have JUKI DLM 5400N in working condition, and instead is using DDL 8700 for collar run stitch, it should be followed by the trimming operation to be done after run stitch by a helper using a pair of scissors or by an overlock machine without thread before the piece can be taken for collar turning. As the additional trimming work content was not a part of the original SAM, the production department should immediately inform Industrial Engineering for amendment of SAM. However, in practice, such incidences are often not reported, with the result that the actual production is lesser than the target and leads to a misunderstanding between the IE and the production department. The Industrial Engineering department could give different machine options and corresponding SAM value to avoid any such misunderstandings. Probably, some of you may have noticed that some machine brands like Duerkopp Adler and Juki often mention two alternative machine types for every operation with the variable expected output. The IE department has to be sensitive to the possible constraints that the production department may face, and accordingly create options with clear ranking as per the most preferred and the least preferred such that the production staff will choose the best choice in any given circumstances. If the best options are not followed for any reason, the production department will be answerable with appropriate reasons. This may result in differential SAM value being applied to the same garment on different days because one particular machine may not be working on one day, but may be working the other day.

GAUGE PARTS CHANGES, ATTACHMENTS AND WORK AIDS

Folders, edge guides and hemmers to be used in an operation are mentioned in the OB, but gauge parts (such as presser foot, feed dog and throat plate) that may change as per the attachment being used, are usually not mentioned. Using attachments without gauge parts can impact the quality and performance of the work aids and lead to rejection/failure of the attachment. The throat plate specification is synchronized with the needle size where the ratio between the needle diameter and needle-hole diameter in the throat plate is maintained to ensure that there is no needle-cut or fabric dip-in. The presser foot can be solid or hinged, single or double leg, narrow or wide, normal or compensating, metal or Teflon. Although there is a separate column in the OB for presser foot, it is often noticed that the word 'standard' is mentioned against every operation. The IE should have a complete idea of the different types of the presser foot and their use, and be able to recommend the most appropriate one. For example, shirt front placket sewing should use the solid presser foot, yoke topstitch should use the left compensating presser foot, and so on.

It is a common practice to fix the attachment in a sewing machine without changing the gauge parts as changing parts take time. Though often written as 'Attachment and Work Aid' within one column in the OB, the IE should, in fact, mention them separately. For instance, the presser foot and the throat plate are machine parts, while a folder and a waterfall stacker are separate devices. Also, an edge guide used for collar tip stitch is an attachment, and the stepped rack used to store the pre-sewn and post-sewn piece is a work aid used in an operation.

NEEDLE TYPE

The OB should mention three different characteristics of the needle: First, what type of needle should be used (e.g. DBX1 for lockstitch, DCX1 for overlock, or UY 128 GAS for flatlock); Second, what point type (e.g. 'R', SES, RG, etc.); and Third, what size (e.g. 90/14 or 75/11). Although, in most of the situations, the needle type is specific to a particular machine type and even if it is not mentioned in OB, there is very less chance of fixing a wrong type in production. In very rare cases like DAX1 and DBX1, both can be used for single-needle lockstitch and appropriate needle type may be mentioned based on the fabric type. Most OBs have one column that mentions the model number of the brand. For example, Groz-Beckert SAN 10 only states that Groz-Beckert is a brand and SAN 10 is the model number and does not give any information about the needle type, point type and size. The complete information should read as Groz-Beckert SAN 10, DBX1, SES, 75/11. So it is advisable that these three important points be mentioned in three different columns.

Generally, for one type of fabric or garment, a single point type of needle is used, but for certain operations, different point styles can be applied, depending on the seam thickness and construction, which will decide the variable. A denim jeans, for instance, should be stitched using a light point needle to avoid any visible material damage after washing, but for operations like Bartacking and Eyelet Button Hole, a sharp point or special ball point like RG should be used to avoid needle buckling and breaking on heavier seams like Bartacking.

SEAM TYPE

The seam type used in certain operations is an important specification in the construction of a garment. Words like 'superimposed seam' and 'flat & fell seam' are familiar, but the nomenclature is not standardized across companies and countries. Therefore, instead of using names, either seam codes like SSa-1 or seam diagram are used. Although the seam codes are internationally standardized, seam diagrams are more commonly used for making representations since they are easy to visually communicate to the sewing operator, and are therefore, the most common methods of representation.

SEWING THREAD

Although it is the buyer who often mentions the thread type to be used, the buyer will rarely mention the different thread types and the thickness that can be used in different operations and also in upper and lower threads of the seam. The IE is required to suggest the most technically appropriate threads for different operations in the OB, for any specific style without changing the aesthetics and commercial value of the product. The selection of thread (for any uncommon style) can often be done in consultation with the thread suppliers.

Providing information on thread type and thread count is important as they can change from one operation to another. Nowadays, thread consumption for different operations can be calculated accurately, and are indicated for controlling/checking the consumption pattern. The thread type and its consumption in different operations determine its quality and procurement for the shopfloor. While the upper thread type (needle) and lower thread type (bobbin/looper) remain the same in almost every operation, some specific operations may require different threads. In such cases, it should be clearly mentioned in the OB. The buyer may simply mention the thread type as '100% polyester' in the techpack; it is the industrial engineers who should add value to the product and process by specifying PPC in the needle and TXP in the looper.

Depending on the technical expertise available with the buyer, these changes often require buyer's approval, but sometimes the buyer may simply trust the manufacturer to use the right thread type. For example, the most suitable and recommended thread combination for knitted T-shirts is a poly core (PPC) spun thread in the needle and texturized polyester (TXP) thread in the looper for the overlock (e.g. Coats Epic is a poly core spun thread, and Gramax is a texturized polyester thread).

Thread tension and presser foot pressure are two important parameters that should be mentioned in the OB to ensure consistent seam quality. As the number of layers of fabric may vary between different operations, the thread tension will also vary to get the perfectly balanced stitch for different stitch types. Similarly, presser foot pressure is very important for getting pucker free seam appearance. It is common practice for floor supervisors to set the thread tension and presser foot pressure by trial and error method. However, thread tension meter (Schmidt, Germany) and presser foot pressure meter (NPT 2010 from Nippo Sewing Machine Company, Japan) are available to objectively measure and mention the numeric value in OB for standardization and repeatability of the performance of the operation.

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INDUSTRY PRACTICES

Are both operation flow chart as well as the Operation Bulletin (OB) prepared in factories? Is it necessary to make a flow chart for every style... or is the OB sufficient? What all information does OB shoot contain? Why OB prepared by IE is not always followed by the production staff? How can this problem be solved? Do Industrial Engineers lack garment assembling technique and machine and attachment know-how?

Flow charts are significant in providing a solution for line planning and section sequencing in an industry which caters to multi-product types that require periodic changeovers. As our unit is product-specific and there is negligible change in the styles, hence we prepare only Operation Bulletin (OB) and a new flowchart is never needed. It just requires one time setting with continuous improvement and update in the process. The OB carries all important information regarding operation-wise manpower allocation, workstations, machines, attachments and targets for each and every operation.

In case of factories doing multiple products, OB and flow chart are essential for every product and style. In such cases, flow chart will give an overview of material flow in lines and the OB will help in balancing the line properly with minimum start-up loss. In the OB, critical operations are highlighted so that they can be taken care of properly with the styles in the production.

There are a number of constraints which do not let production staff follow the OB, such as availability of skilled and multi-skilled operators, availability of equipment, poor maintenance management and conventional mentality of production staff. At supervisory level and below, people are reluctant to change. They prefer working on the basis of their experience in spite of referring to the technicalities and specifications provided by the IE department. For instance, sometimes supervisors don't prefer increasing the number of workstations in order to improve or maintain floor efficiency. This problem can be solved by maintaining and updating an operator's skill matrix on a regular basis and following a good maintenance system, preferably of preventive maintenance, which will never result in deficiency of workstations.

SANJAY ANAND
Executive Director, Mohan Clothing Co. Pvt. Ltd. (Blackberrys), Gurgaon

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Our suppliers prepare the operation breakdown flow chart as well as the Operation Bulletin (OB) and both the documents are prepared for specific purposes. While an OB provides us with an accurate requirement of machines and manpower, the operation breakdown flow chart is a crucial visual tool for ensuring that the product, process and material are moving for maximum productivity without any redundancies or unnecessary material and/or manpower travel. Besides complete information of machines, attachments and consumables, the OB also mentions the expected target, seam diagrams and SPI.

INDRA BHARDWAJ
Production Director, Continuum Buying Agency, Gurgaon

As an ISO 9000 compliant unit, both Operation Bulletin (OB) and flow chart are vital for our output and quality. The operation breakdown and flow chart keep track of all the iterations provided by the IE, QA and production managers such as the OB documents machines, attachments and consumables details. But they do not document seam diagrams, mainly because it can create avenues of misinterpretation as most of the times, amendments keep coming in till the last minute and it becomes difficult and time consuming and the rewards are not equivalent either. Besides it is time consuming for small order-runs and the operators understand practical demonstrations better than seam diagrams. We do not put thread consumption also in the OB as it is not of interest to the production manager. Basically, whatever one puts on, an OB should contribute to productivity and not promote confusion.

MANJUNATH P
Technical Manager (Knits and Woven), Go Go International, Bangalore

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Operation Bulletin (OB) is a very important and integral part of the manufacturing process and works well for large order-runs and where line systems are followed. Production meets are important and appropriate occasions to explain the OB, as it ensures that each operator knows the nitty-gritty of the operation process to be followed. This implies standardization and goes a long way in controlling the quality of the production at needle point and reduces the chances of re-works. Most of the production staff view IE as a hindrance because IE breaks down each operation of a garment and there is no scope of excuse for production loss or lapse. The issue can be resolved by empowering the IE department.

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